README

Supporting Research Data for Building Energy Flexibility Assessment in Zero-Carbon Buildings

Related manuscript: *Assessing Building Energy Flexibility in a Next-Generation Zero-Carbon Building: An Integrated SFS-CRITIC-DEMATEL and GRA-TOPSIS Framework*

1. Overview

Study cases	Qingdao, Beijing, Nansha and Nanjing.
Evaluation method	Spherical fuzzy sets (SFS), CRITIC–DEMATEL combined weighting, and GRA–TOPSIS.
Data coverage	Indicator system, expert evaluations, weights, selected calculation results, four-case BEFI values, weight-sensitivity results and hypothetical operational-improvement scenarios.
Data sources	Related manuscript (Tables 3 and 5–10; Appendices A–F), with detailed sensitivity values from the accompanying calculation workbooks.
Expert labels	E1–E5 denote the five anonymized linguistic evaluation columns used for each case.
SFS components	Membership, non-membership and hesitancy are the three components of each spherical fuzzy value.

2. Computational workflow

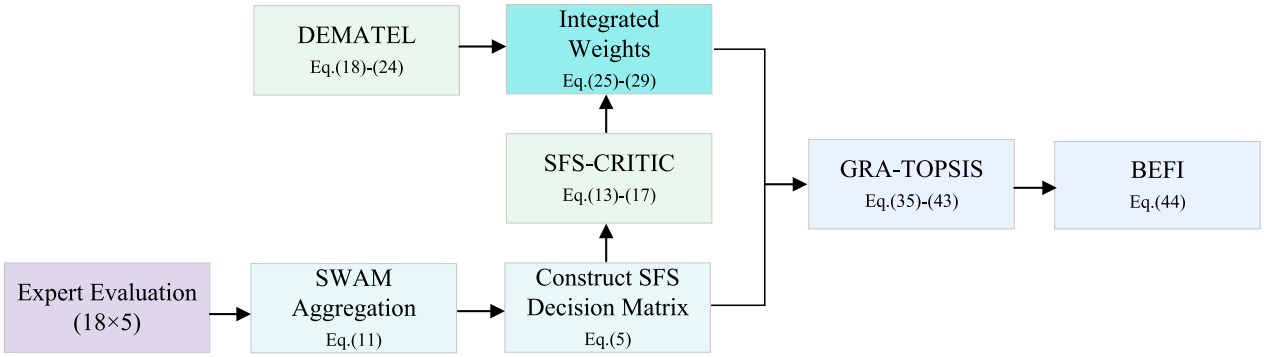


Figure 1. Computational workflow of the SFS-CRITIC-DEMATEL and GRA-TOPSIS framework. Equation numbers refer to the manuscript.

3. Data files

File	Content	Workflow step (Figure 1)	Manuscript source
`01_Indicators.xlsx`	Codes, names and primary dimensions of 18 secondary indicators	Indicator system	Table 3
`02_Rating_Scale.xlsx`	Nine linguistic ratings and their spherical fuzzy values	Expert evaluation	Table 5
`03_Expert_Ratings.xlsx`	Five expert ratings for each of 18 indicators in four cases (360 ratings)	Expert evaluation (18×5)	Appendices A and F.1
`04_Indicator_Weights.xlsx`	Subjective, objective and combined weights for Qingdao; objective and combined weights for Beijing, Nansha and Nanjing	SFS-CRITIC, DEMATEL, integrated weights	Table 6 and Appendix F.2
`05_Qingdao_SFS.xlsx`	90 weighted spherical fuzzy values for the Qingdao case	Weighted SFS decision matrix	Appendix B
`06_Qingdao_Ideals.xlsx`	Positive and negative ideal solutions for Qingdao	GRA-TOPSIS	Appendix C
`07_Qingdao_Grey.xlsx`	Grey relational coefficients for Qingdao relative to positive and negative ideal solutions	GRA-TOPSIS	Appendices D and E
`08_Case_Results.xlsx`	Four-case BEFI values, five expert-level relative closeness values per case, flexibility levels and main resources	BEFI	Tables 7 and 9; Appendix F.3
`09_Sensitivity_28.xlsx`	28 BEFI results across four primary dimensions (J1–J4) and seven weight-change levels (−30%, −20%, −10%, 0%, +10%, +20%, +30%)	BEFI (sensitivity analysis)	Section 6.1 and Figure 8
`10_Scenarios.xlsx`	Qingdao baseline and four hypothetical operational-improvement scenarios (S0–S4)	BEFI (scenario analysis)	Table 10

4. Notes

- **Numerical precision.** Reported values follow the manuscript's displayed precision; sensitivity calculations retain additional decimal places. Small differences in weight sums can result from rounding.
- **File format.** All data files are Microsoft Excel workbooks (.xlsx); the first row of each file contains the column headers.